

Yun-Tung (Winona) Chu

yuntungchu.ytc@gmail.com | (669) 609-4418 | Seattle, WA | winona1111.github.io | linkedin.com/in/yuntungc/

EDUCATION

University of Washington

Seattle, WA

M.S. in Electrical and Computer Engineering | GPA: 3.86 / 4.0

Expected June 2026

Relevant Coursework: Distributed Systems, AI For Mobile Robots, LLMs, AR/VR, SW Engineering in Embedded App

National ChengChi University

Taipei, Taiwan

B.S. in Management Information Systems | GPA: 4.04 / 4.3

January 2024

Relevant Coursework: Data Structure, OS, Database Management Systems, Algorithms (Computer Science)

PROFESSIONAL EXPERIENCE

Software Engineer Intern

May 2026 - Present

Khorium (NVIDIA Inception startup): Remote AI-powered CFD/FEA simulation

Remote, US

- Engineered full-stack fix for a 1.6x overestimation in mesh cost prediction: replaced bounding-box surface area proxy with OCC-derived CAD face areas in Python backend, piped through PostgreSQL JSONB to a Zustand-managed React/ TypeScript frontend, and recalibrated runtime model from empirical production data.
- Unblocked cross-repo deployment failures in a multi-service AWS CDK pipeline by diagnosing a broken Alembic migration and resolving a cascading generated-type conflict.

Fullstack Software Engineer

Jan 2025 - June 2025

astrolabe analytics (UW Capstone)

Seattle, WA

- Built a multimodal data ingestion platform with React and FastAPI, exposing 15+ RESTful endpoints to ingest, standardize, and query battery datasets across 6 file formats, backed by AWS RDS (PostgreSQL) and S3. [Web]
- Developed LLM-powered client-server data standardization pipeline using OpenAI API for semantic schema alignment, increasing the column matching rate to 90%.

Software Development Engineer Intern

July 2022 - Aug 2022

Innodisk Corporation

Taipei, Taiwan

- Built a real-time screw welding anomaly detection pipeline targeting edge deployment, expanding the labeled dataset from 1.3k to 7.6k training images (~5.6x) via geometric and noise-based augmentation.
- Fine-tuned YOLOv4-tiny using a Keras-based deployed to Xilinx KV260 via Vitis-AI (.h5 → q.pb → .xmodel), achieving 100+ FPS at 8W on DPU with 90% mAP@0.45 post-quantization (5% drop from int8 quantization).

PROJECTS

Machine Learning Research Assistant at Information Processing Lab (UW)

July 2025 - Present

- [CVPR 2026 Workshop] First-authored research on zero-shot single-temporal anomaly detection in remote sensing using CLIP + DINOv3 dual-vision fusion (BSSRS), achieving 95.38% Pixel AUC.
- Fine-tuned a lightweight Qwen3.5-VL-4B with LoRA on MixedWM38 (38k+ images) for semiconductor defect VQA, optimizing the model footprint for high-throughput inference; applied GRPO with custom reward functions to mitigate hallucinations and resolve multi-label combinatorial explosion.
- Architected scalable distributed training and inference pipelines on HPC clusters including Taipei-1 (NVIDIA DGX SuperPOD) and NCHC; integrated Flash-Attention 2 and vLLM to maximize PagedAttention and VRAM utilization and avoiding OOM for high-throughput multimodal workflows.
- Developed end-to-end auto-annotation pipeline (GroundingDINO + SAM2), generating 50K+ labeled images and reducing manual annotation effort by 50%.

Sharded Key-Value Distributed System

Sept 2025 - Dec 2025

- Architected a fault-tolerant, linearizable key-value storage system in Java using Multi-Instance Paxos for consensus and log replication, ensuring data consistency across server failures and network partitions.
- Implemented a Shard Master for zero-downtime dynamic re-sharding and engineered Two-Phase Commit (2PC) transactions to guarantee ACID properties.

HoloLights: Color Vision Deficiency Assistance Software Module

April 2022 - Dec 2023

- 1st Place Winner, 2023 National Universities Innovation Competition: Developed a C#/Unity MR app on HoloLens 2, optimizing socket latency by 100ms for real-time edge-server synchronization.
- Integrated TensorFlow Object Detection (94% mAP) and custom color transformation algorithms to assist users with color deficiencies. [Github]

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, C#, Bash, PowerShell, SQL

Web and Backend: React, TypeScript, HTML/CSS, RESTful APIs, FastAPI, Postman, Node.js

Database and Cloud: MySQL, MSSQL, PostgreSQL, Amazon Web Services (AWS) S3, Git, GitHub, GitLab, Docker

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, HuggingFace, SLURM, vLLM, Flash-Attention2, HPC